UNIVERSITÉ DE FRIBOURG
UNIVERSITÄT FREIBURG

WORKSHOP FRIBOURG 2026

INTRODUCTION TO METAGENOMICS DATA ANALYSIS OF MICROBIAL COMMUNITIES

Laurent Falquet
Jeferyd Yepes-Garcia

Swiss Institute of Bioinformatics

Modalities of the workshop

Each topic is organised in 1h theory + 2h exercises (see moodle page)
https://moodle.unifr.ch/course/view.php?id=317449

or https://moodle.unifr.ch and search for
« SIBMetagenomicsWorkshop26 »

Key: SIB_Metagenomics26

Coffee breaks (near the teaching room)
Lunch breaks not included, but two affordable cafeteria nearby.
Social Apero this evening (5:30pm).

Program of the workshop

Day	Topic
DAY1am	**General Introduction**, goals & opportunities **Intro to NGS and QC** of the reads
DAY1pm	**Taxonomic classification**, contaminant removal
DAY2am	**Whole Genome Shotgun sequencing part 1** Comparing Pipelines & workflow managers (i.e SnakeMake, Nextflow)
DAY2pm	**Whole Genome Shotgun sequencing part 2** Downstream analyses, gene prediction prokka, annotation, function, pathways, eggNOG, KEGG, AntiSmash secondary pathways in bacteria
DAY3am	**Targeted sequencing** ASV vs OTU
DAY3pm	**Targeted sequencing** Downstream analysis

Who am I ?



Laurent Falquet
Swiss, born in Geneva

PhD in Biochemistry at UniGe
Post-doc at the SIB Swiss Institute of Bioinformatics
Responsible of the Swiss EMBnet node
Project manager at Vital-IT High Performance Computing center

Since 2013
Senior Lecturer manager of the Bioinformatics core facility at UniFr, Switzerland
Group leader of the Swiss Institute of Bioinformatics

Microbiome amplicon projects: past

Exploring the phyllosphere of plants resistant to pathogens

(Arabidopsis thaliana, leaf surface, 16S)

Evaluating items of scientific findings in cases where the Defense says: 'It is my twin brother'

(human saliva, 16S, rpoB)

Involved in data analysis of the PromESSinG:

Promoting EcoSystem Services in Grapes https://www.promessing.eu/

(soil, 16S, ITS)

Targeted vaccination of the plant microbiome for sustainable crop protection:

SOLANUM (SOiL ANtifungal vaccination Using native Microbiome) **with Prof. Laure Weisskopf, UniFr (soil & leaves, 16S)**

Metagenomics analysis of mother & infant gut microbiome.

(gut, milk & saliva, WGS) HFR, Fribourg

Metagenomics projects: current



Started in September 2022:

Developing high-throughput strategies for efficient rice straw degradation through metagenomics data analysis and network reconstruction of metabolic pathways by deep learning approaches. (metagenomics, rice straw & soil, WGS)

A collaboration with Prof. D.Uribe and Prof. E.Barreto, Colombia

Started in September 2023:

soil FL 16S + WGS, tomato and wheat rhizosphere

A collaboration with Dr. Bilal Rahmoune, Algeria

Started in 2025: **FL 16S and ITS of cow slurry** (Swiss company)

Other: MC member COST Action Machine Learning for Microbiome (finished) and WG member COST Action MiCropBiome

Your turn!

Please introduce yourself and your interest in metagenomics



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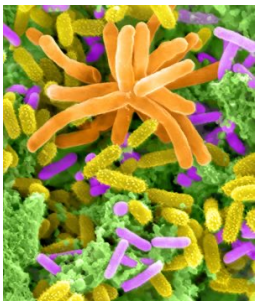
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WHAT IS METAGENOMICS?

Laurent Falquet
Jeferyd Yepes-Garcia

SIB
Swiss Institute of Bioinformatics

What is metagenomics?



Methylamine-enriched community of Lake Washington sediment featuring Methylotenera cells Photo © Dennis Kunkel Microscopy, Inc. (Color by Ekaterina Latypova)

"The study of the DNA of uncultured organisms"

> 99% of all microbes do not grow *in vitro*

Some definitions…

A genome:

Entire genetic information of a single organism

A metagenome:

Entire genetic information of an assemblage/mixture of organisms

A metatranscriptome:

Entire collection of transcripts of an assemblage/mixture at a given time

https://en.wikipedia.org/wiki/Metagenomics

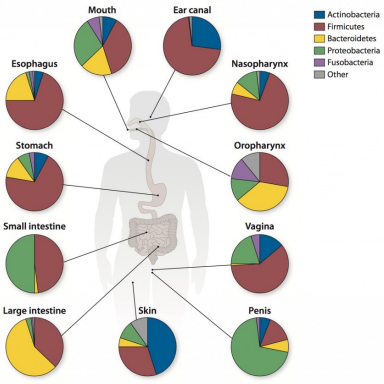
What is a microbiome?



The microbiome (or microbiota):

is an ecological community of commensal, symbiotic and pathogenic microorganisms found in and on all multicellular organisms studied to date from plants to animals. A microbiota includes bacteria, archaea, protists, fungi and viruses.

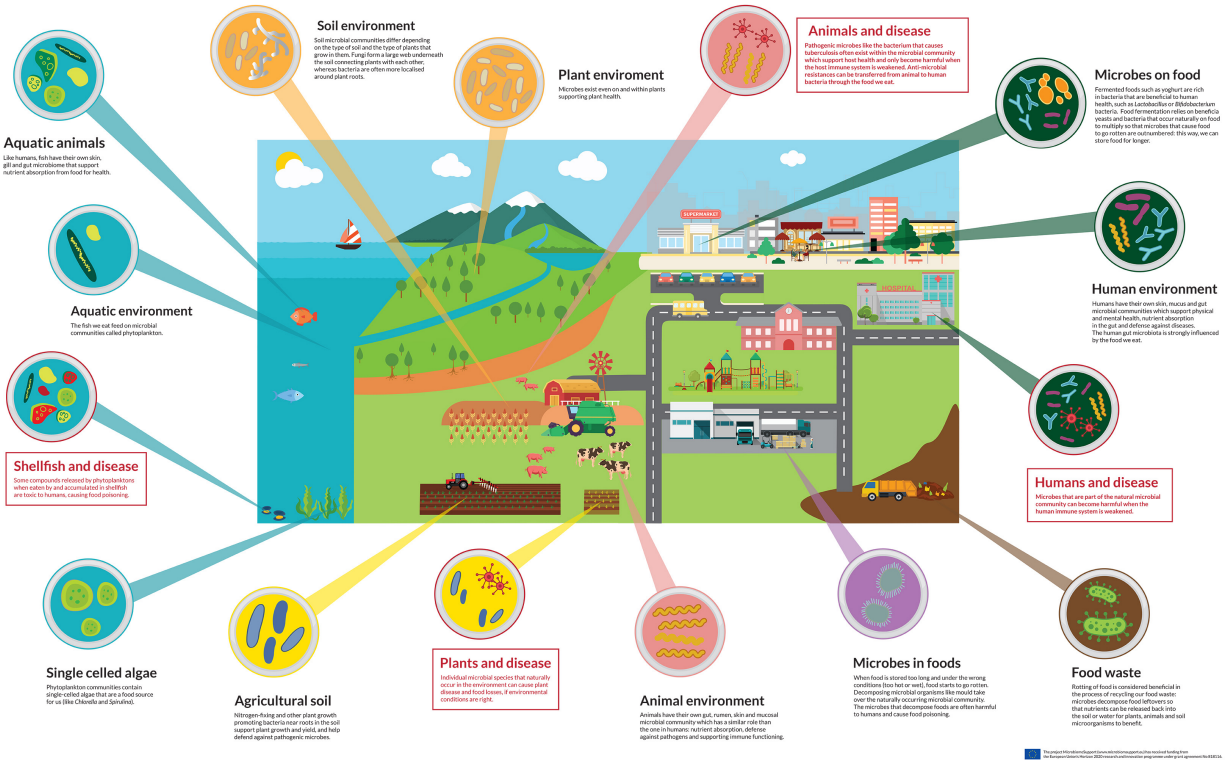
https://en.wikipedia.org/wiki/Microbiota



⚠ **Flora or microflora** synonym should not be used in scientific literature. It can be found in popular literature or in a yogurt/probiotic advertisement destined to the general public.

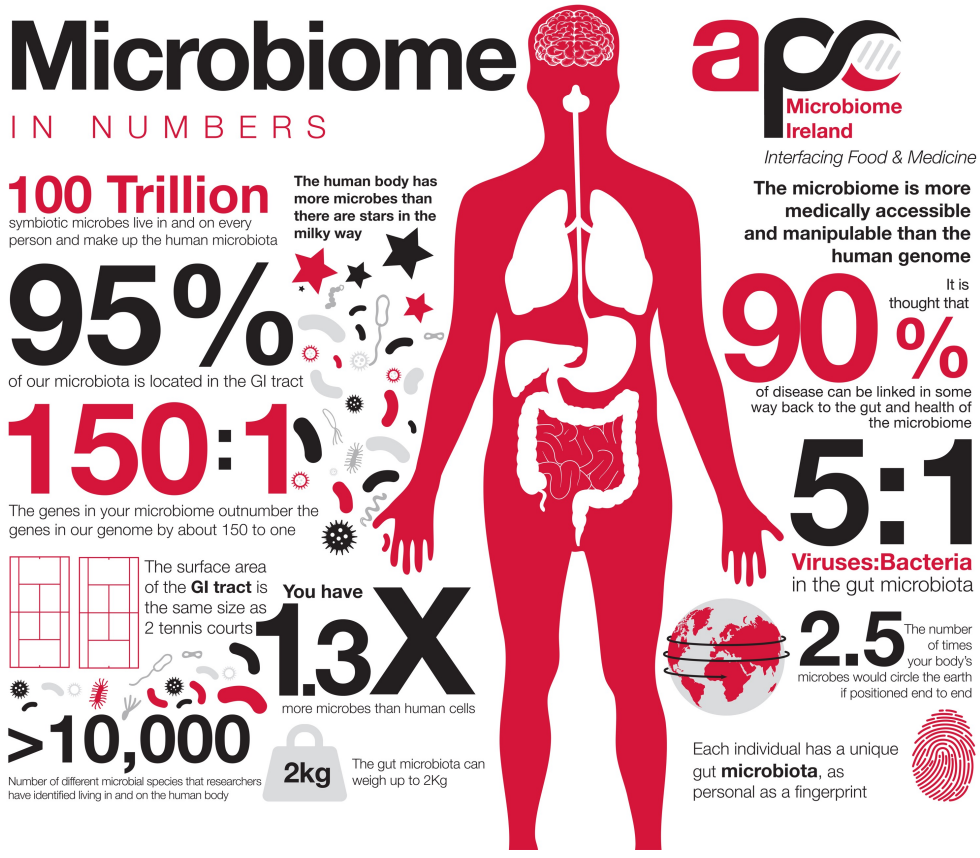
https://web.archive.org/web/20221205041830/https://owlcation.com/stem/The-Ecology-of-the-Human-Microbiome

Microbes are everywhere



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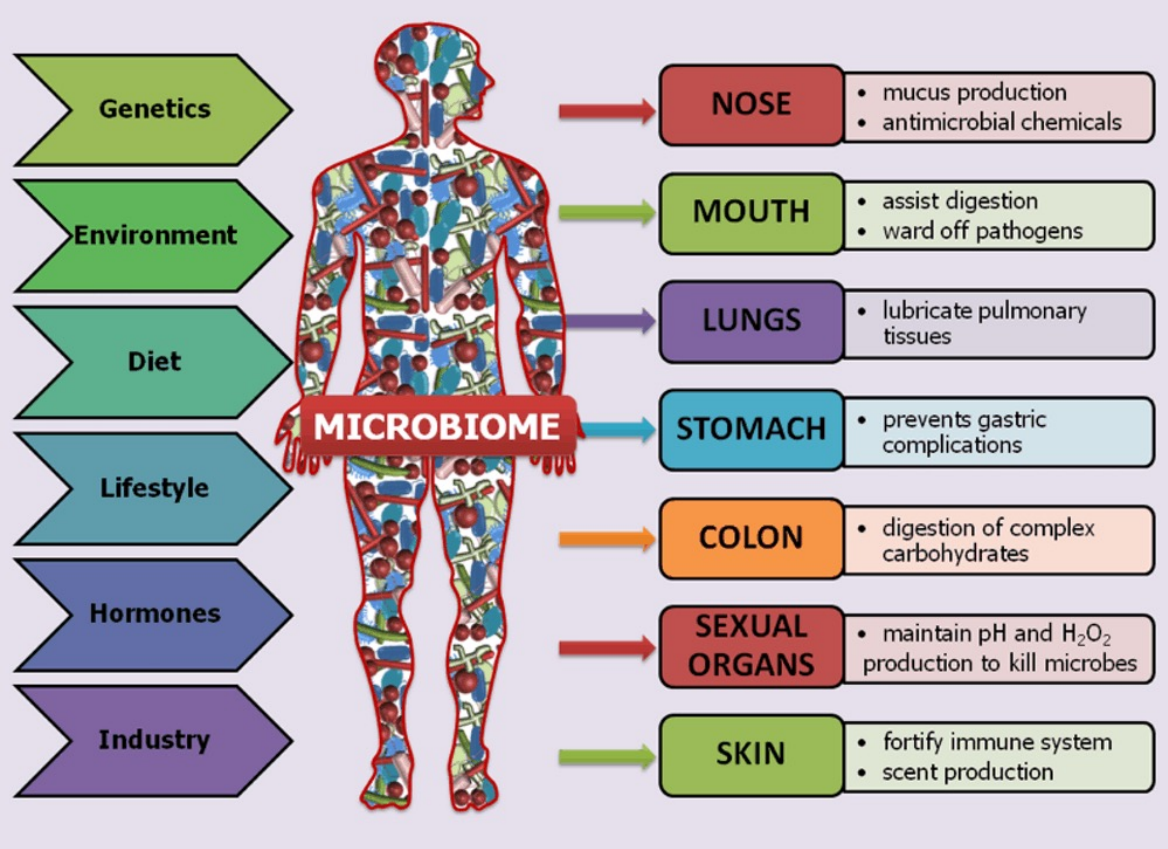
https://www.microbiomesupport.eu/microbes-are-everywhere-in-the-foodsystem/



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https://worldmicrobiomeday.com/wp-content/uploads/2019/06/2019-Microbiome-in-Numbers-1.jpg

Microbes are everywhere on us



https://theconversation.com/the-human-microbiome-is-a-treasure-trove-waiting-to-be-unlocked-118757

Examples of roles and impact of the microbiome in the human gut

Benefits

- digestion, production of vitamins & antioxidants, iron absorption, fighting pathogens

Diseases

- obesity, diabetes, some cancers, heart disease

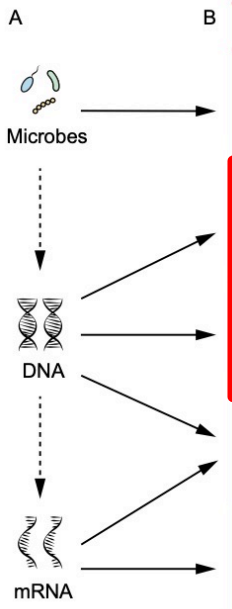
Immunity

- eczema & allergy, asthma, autoimmune disorders

Behavior

- psychiatric disorders (depression), autism

How to study a microbiome/metagenome?



Method	Advantages	Limitations
Culturome	• High-throughput • Targeted selection • Provides microbial isolates	• Expensive • Laborious • Influenced by media and the environment
Amplicon (16S/18S/ITS)	• Quick analysis • Low-biomass requirement • Applicable to samples contaminated by host DNA	• PCR and primer biases • Resolution limited to genus level • False positive in low-biomass samples
Metagenome	• Taxonomic resolution to species or strain level • Functional potential • Uncultured microbial genome	• Expensive • Time-consuming in analysis • Host-derived contamination
Virome	• Can identify RNA and DNA viruses • Quick diagnosis	• Most expensive • Difficult to analysis • Severe host-derived contamination
Metatranscriptome	• Can identify live microbes • Can evaluate microbial activity • Transcript-level responses	• Complex sample collection and analysis • Expensive and complex in sequencing • Host mRNA and rRNA contamination

Protein Cell 2021, 12(5):315–330
https://doi.org/10.1007/s13238-020-00724-8

The choice of the method depends on 3 main questions



Who is there? Taxonomic profiling

catalogue of species, diversity, genus, etc.
distribution (how many of each)

What are they doing? Functional analysis

genes/proteins, GO terms, metabolic pathways
functional annotation, bioprospection



How do they compare? Differential analysis

pairwise or multiple comparisons
correlation with environmental factors

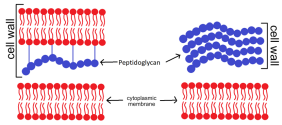


In addition: Meta-analysis

combining multiple studies

What is challenging?

- easy + hard



Lab challenges:

Different DNA extraction steps can give different sequence sets

Data analysis challenges:

Huge amount of sequences to process

Most sequences are absent from the existing reference databases

Assembly of communities can be hard with short reads

No perfect tool or workflow for analysis exists

Different analysis tools can give different results

Even worse, the same version of a tool can give different results depending on the reference database used

Technical options for metagenome/microbiome sequencing

Long reads vs short reads?

Read type	Advantage	Disadvantage
Short (Illumina)	Cheaper to produce More sensitive	Many biases
Long (Nanopore)	Longer sequences	More expensive High error rate
Long (PacBio)	Longer sequences HiFi reads	Less counts

Targeted vs WGS

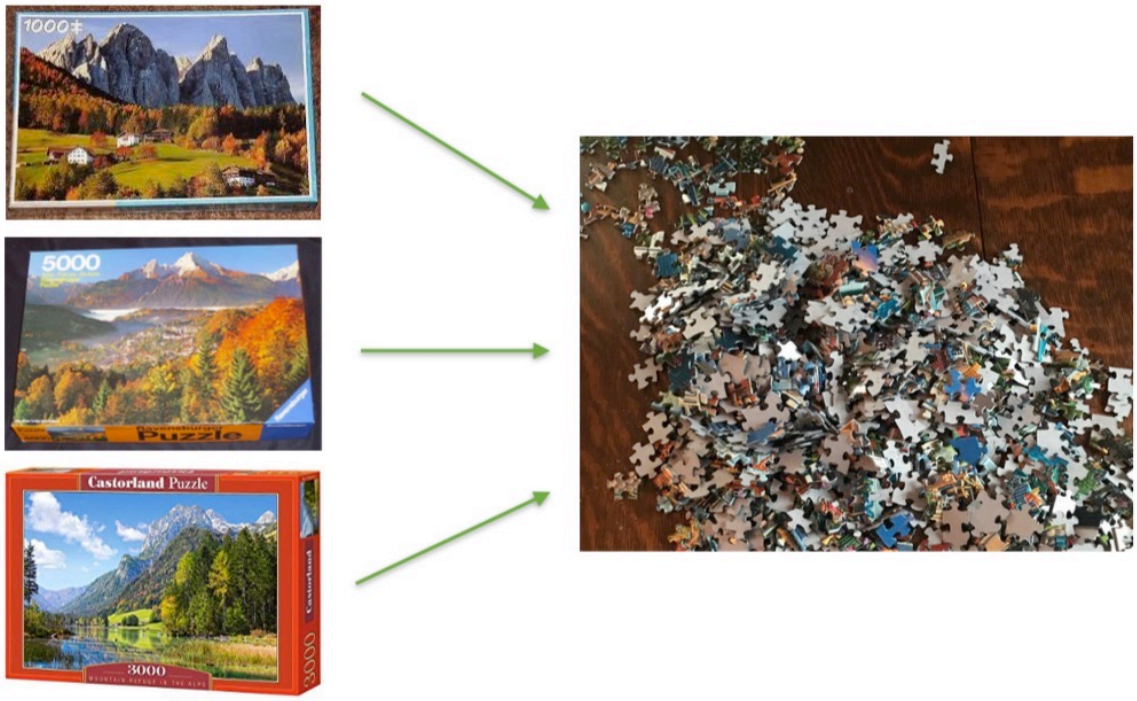
Amplicon (targeted)	WGS
Sequence alignment to a reference database is sufficient	Requires *de novo* assembly
PCR selection of a gene of interest	No selection
16S or ITS or COI or other…	all DNA is sequenced (contaminants!)
Error correction (denoising)	QC/cleaning + *de novo* assembly
Clustering to ASV or OTU	Binning to MAG
Taxonomic assignment	Taxonomic assignment
No function available	Functional assignment
Downstream analysis	Downstream analysis
E.g., alpha and beta diversity	E.g., comparative genomics

Metagenomics via Whole Genome Shotgun (WGS)

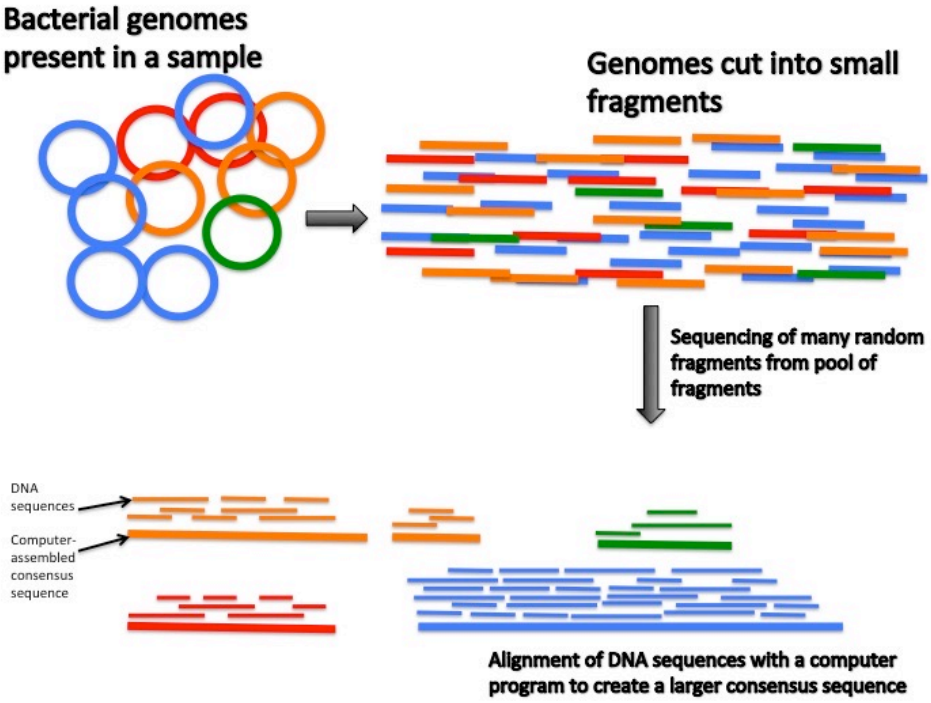
Difficulties:

- **Requires *de novo* assembly**
- **Many genomes (hundreds to thousands)**
- **Often closely related**
- **Coverage very different (abundant vs rare)**
- **Large amounts of data**
- **Host contamination**

Like thousands jigsaw puzzles mixed together!!



Whole metagenome assembly



After genome assembly

- Most metagenomics assemblers use de Bruijn graphs
- Even single genome assemblies are often still highly fragmented
- Metagenome assemblies are worse as many methods used for reducing the fragmentation cannot be applied in a mixture of genomes

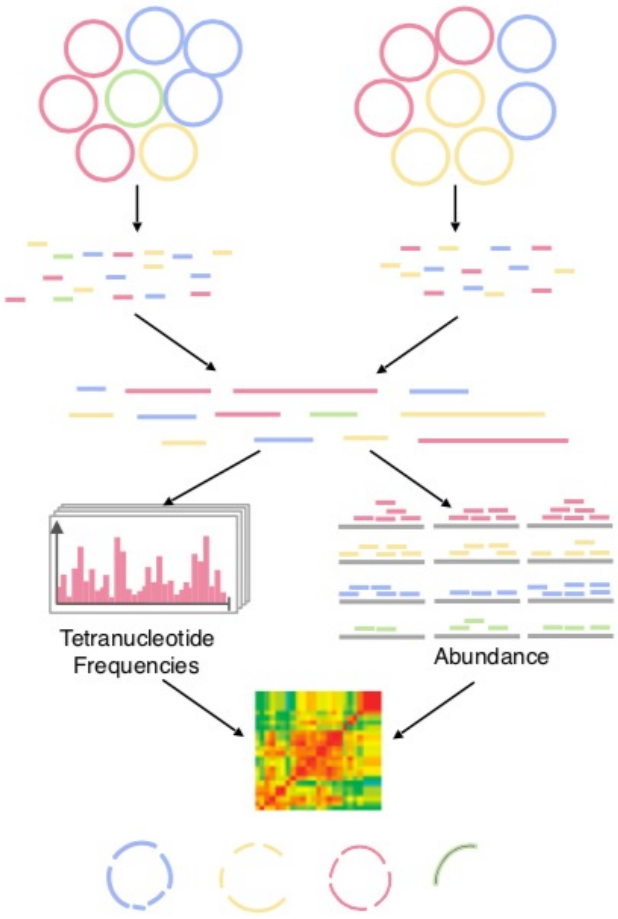
- The idea is to sort the data into smaller groups or « bins »



Assembly and binning

Binning can use various data to produce Metagenome Assembled Genomes (MAGs)

1) Abundance (coverage)
2) Tetranucleotide frequencies

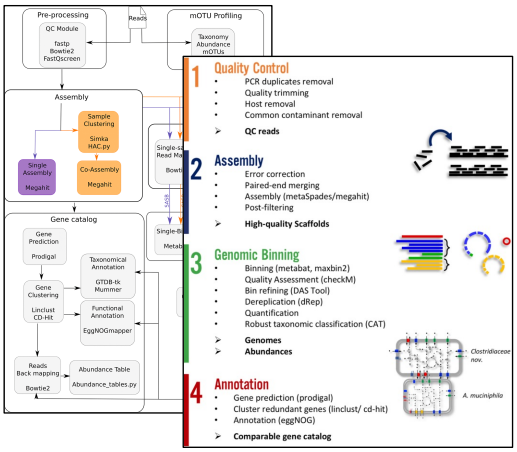


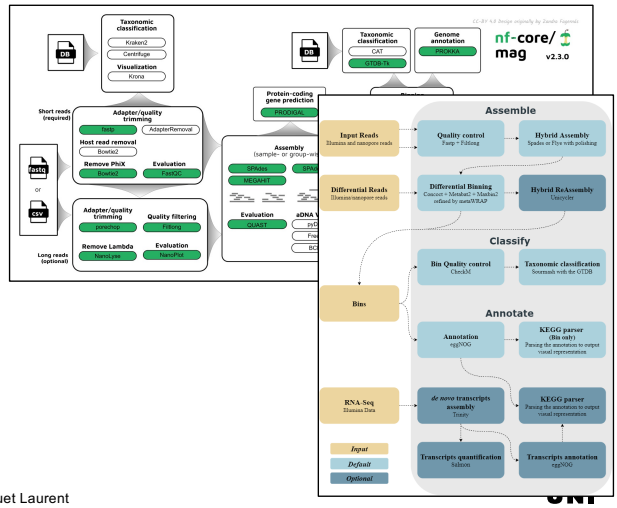
Complete automated workflows and pipelines!

Most use Snakemake or Nextflow or custom pipelines

snakemake

nextflow





Automated workflows and pipelines differences

Many pipelines are currently in development
They usually start from raw short reads or contigs and go to various levels of analysis (mainly to Binning into MAGs).
Some pipelines go even further to annotation of gene catalogs.

MGnify	https://www.ebi.ac.uk/metagenomics/pipelines/5
MG-RAST :	https://www.mg-rast.org
MOCAT2 :	https://mocat.embl.de/about.html
Metagenome-Atlas	https://metagenome-atlas.readthedocs.io
nf-core/mag	https://nf-co.re/mag
MUFFIN	https://github.com/RVanDamme/MUFFIN
METAWRAP	https://github.com/bxlab/metaWRAP
MAGNETO	https://gitlab.univ-nantes.fr/bird_pipeline_registry/magneto

Many others…

The differences are mainly in the tools and databases used, but more importantly the strategies for assembly (co-assembly) or binning (co-binning) or both.

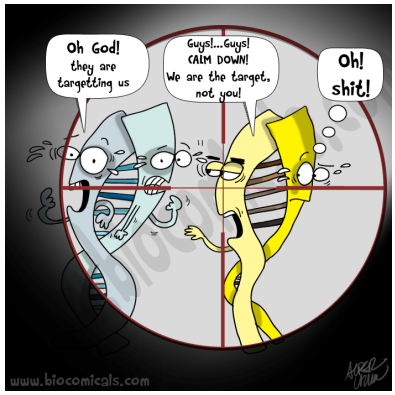
Most pipelines are designed for short reads!

Targeted vs WGS

Amplicon (targeted)	WGS
Sequence alignment to a reference database is sufficient	Requires *de novo* assembly
PCR selection of a gene of interest	No selection
16S or ITS or COI or other…	all DNA is sequenced (contaminants!)
Error correction (denoising)	QC/cleaning + *de novo* assembly
Clustering to ASV or OTU	Binning to MAG
Taxonomic assignment	Taxonomic assignment
No function available	Functional assignment
Downstream analysis	Downstream analysis
E.g., alpha and beta diversity	E.g., comparative genomics

Targeted or Amplicon sequencing

Library preparation requires PCR amplification
QC
Clean reads
optional: combine overlapping PE
Align reads to reference database
Clusterize reads to OTUs
Assign OTUs to taxonomy



Tools (examples):

Mothur
Qiime2
DADA2
MEGAN
Metaphlan
PIPITS (for fungi)
...

Choosing the target of PCR amplification

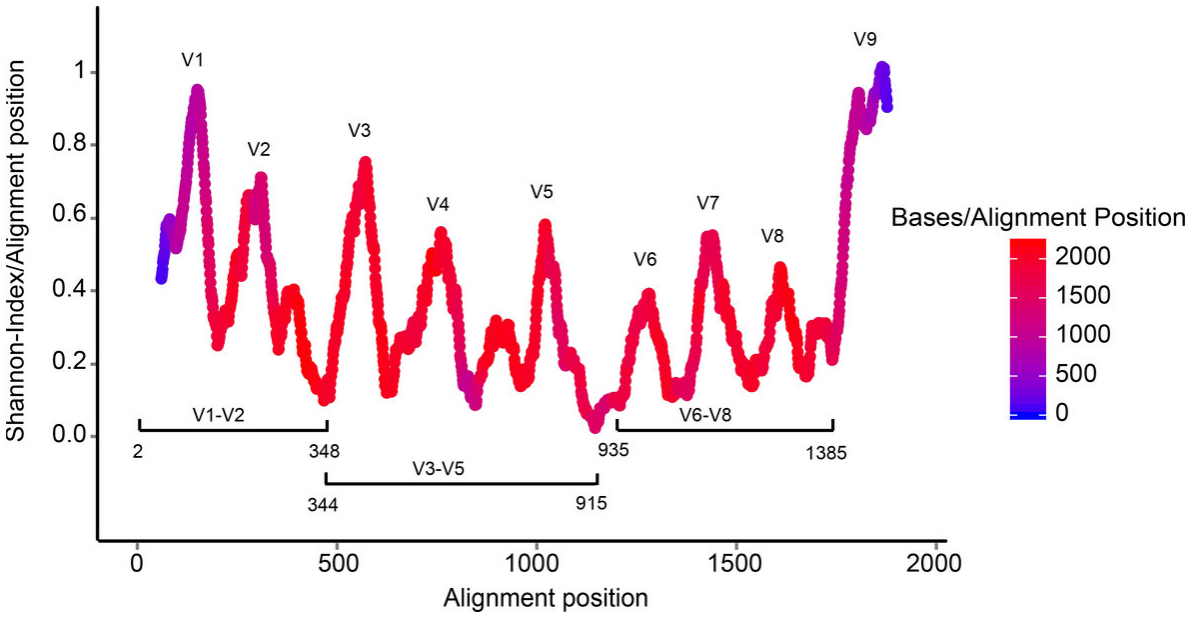
Usually this is a gene with some variable sequences in all species surrounded by conserved sequences:



Several genes have been used : 16S, 18S, ITS, gyrB, rpoB, recA etc.

The most popular being the 16S for bacteria.

Selection of primers: reminder about 16S rRNA variable regions



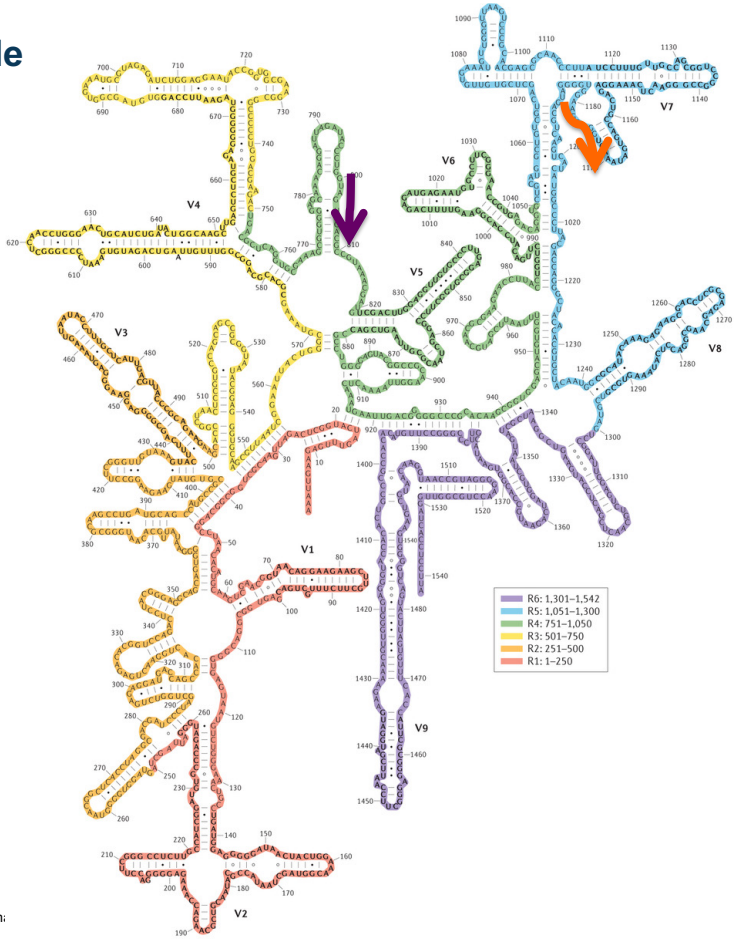
Selection of the primers example

16S rRNA

primers:
799F
1193R
distance ~400bp

Sequenced by MiSeq
2x 300bp -> overlap!

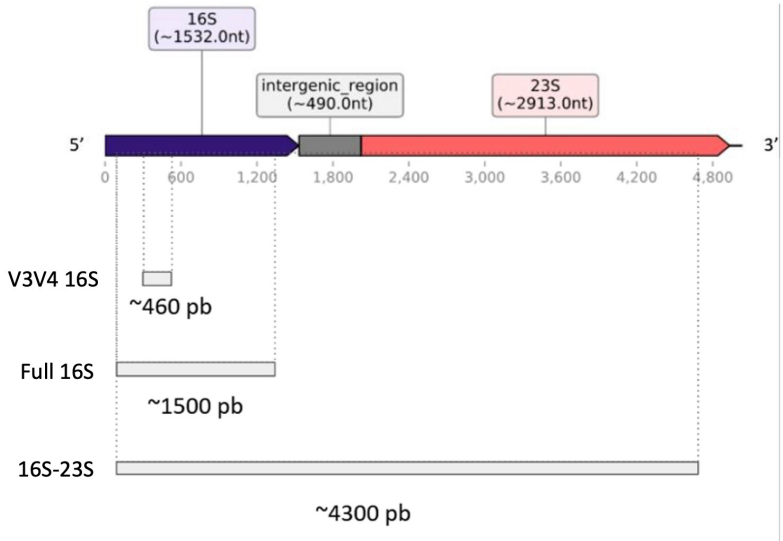




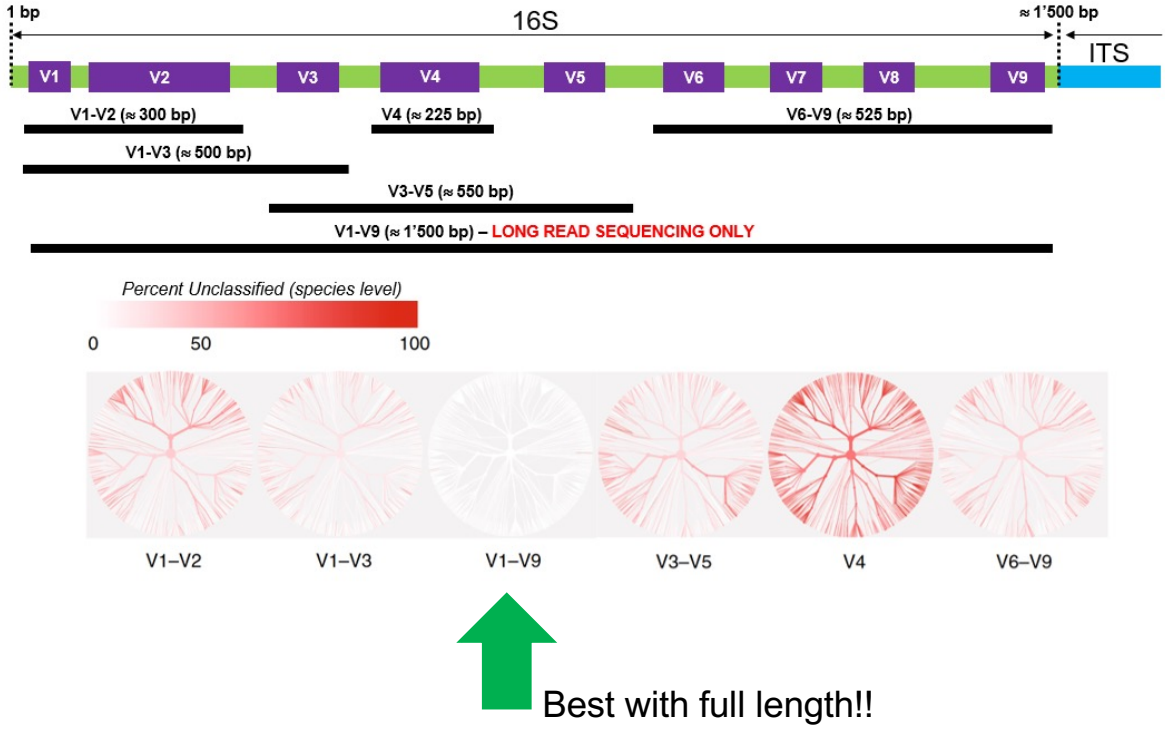
Nature Reviews | Microbiology

Full length amplicon sequencing

With the improvement of long read sequencing and the development of other long read techniques like LoopSeq, it becomes possible to sequence full length 16S or even larger amplicons

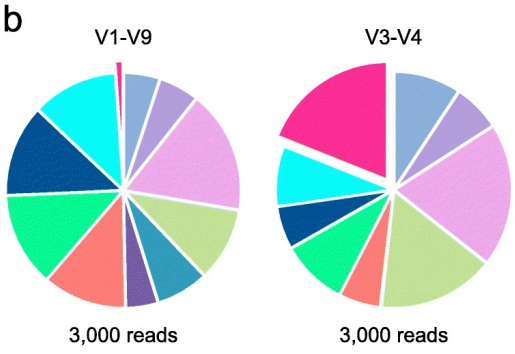


Improved classification with full length 16S

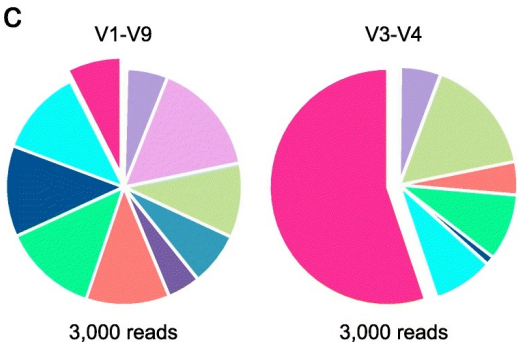


Best with full length!!

Example MOCK analysis with full length 16S *vs* V3-V4 (10 strains even mix)



Genus	V1-V9	V3-V4
Bacillus	5.07%	9.30%
Bifidobacterium	5.63%	6.60%
Clostridium	16.97%	19.70%
Deinococcus	10.23%	16.10%
Enterococcus	7.30%	0.13%
Escherichia	4.50%	0.00%
Lactobacillus	11.63%	5.77%
Rhodobacter	12.97%	9.17%
Staphylococcus	12.73%	5.97%
Streptococcus	11.87%	8.30%
mis- or unclassified	1.10%	18.97%

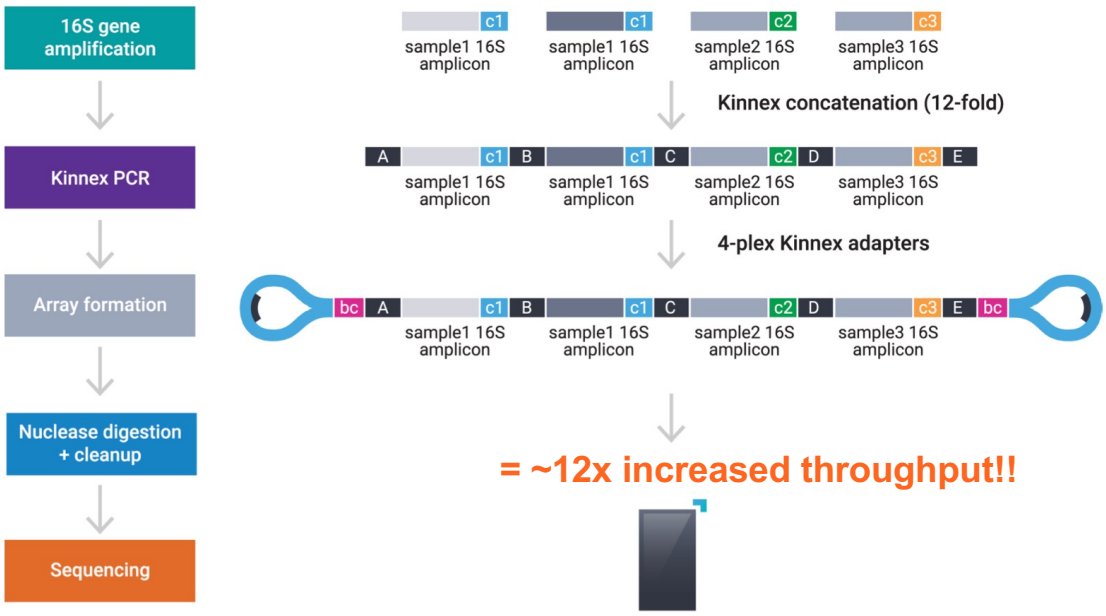


Species	V1-V9	V3-V4
B. cereus	0.40%	0.17%
B. adolescentis	5.63%	5.53%
C. beijerinckii	15.67%	0.03%
D. radiodurans	10.23%	16.07%
E. faecalis	7.30%	0.00%
Escherichia spp.	4.50%	0.00%
L. gasseri	11.47%	4.57%
R. sphaeroides	12.97%	9.07%
S. epidermidis	12.50%	1.20%
S. mutans	11.87%	8.20%
mis- or unclassified	7.47%	55.17%

Matsuo et al. BMC Microbiology (2021) 21:35 https://doi.org/10.1186/s12866-021-02094-5

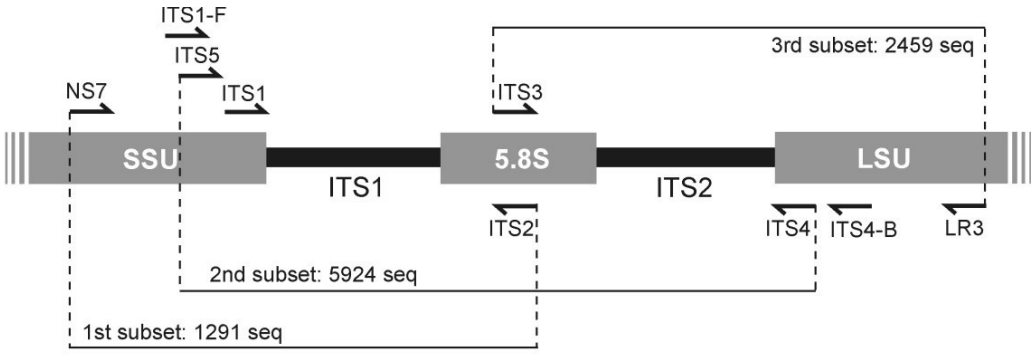
With PacBio: Kinnex 16S rRNA library

Combining multiple full length 16S (up to 12) in one HiFi read!



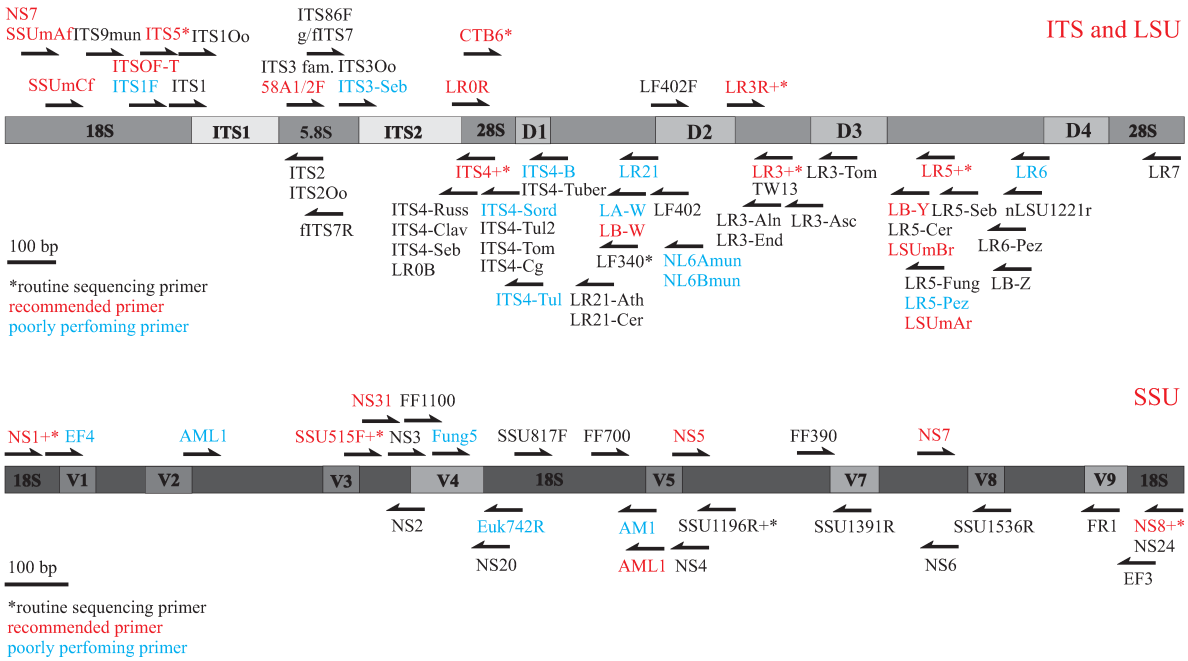
ITS

For fungi, the most widely used fungal genetic markers are the internal transcribed spacers (ITS1 and ITS2) of the nuclear ribosomal subunit, which are located between the small and large subunit gene (SSU/18S and LSU/28S, respectively).



Bellemain, Eva et al. “ITS as an environmental DNA barcode for fungi: an *in silico* approach reveals potential PCR biases.” BMC Microbiology (2010).

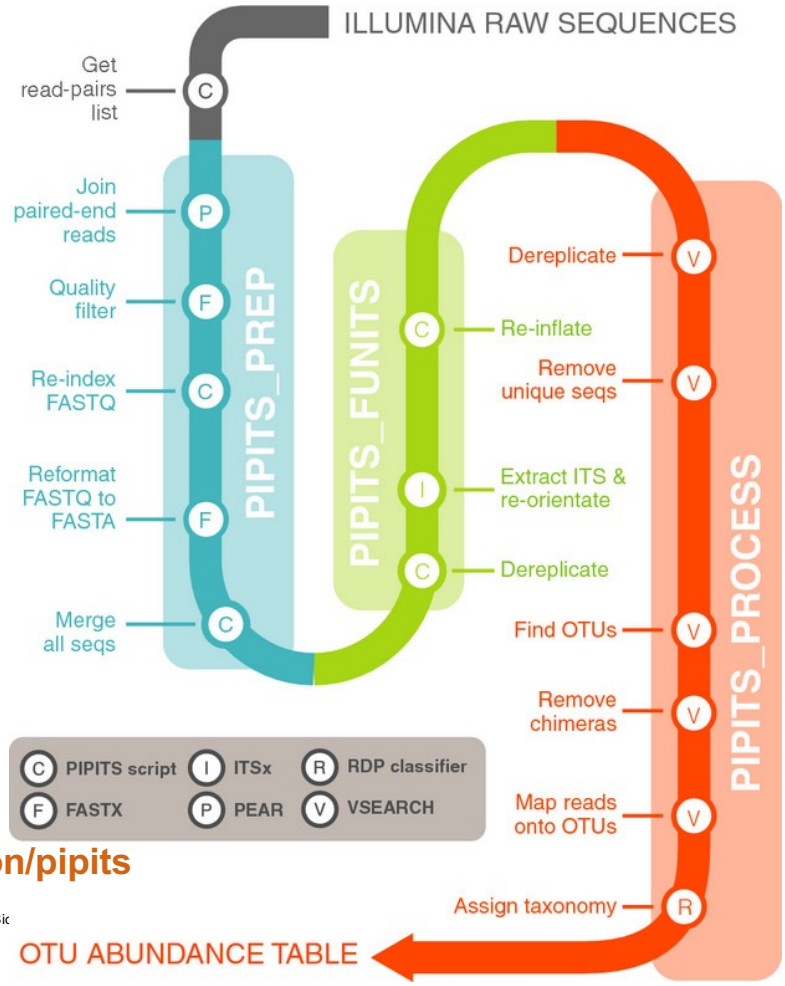
List of primers for ITS and 18S: see UNITe web site https://unite.ut.ee/primers.php



ITS pipelines

PIPITS3

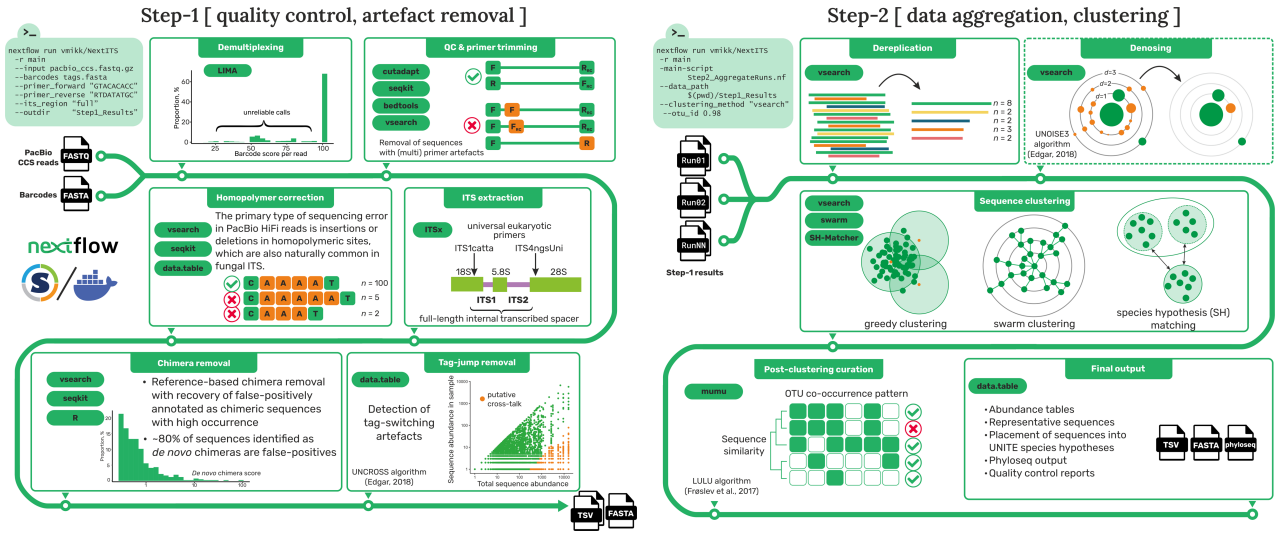
A 3 steps pipeline generating OTUs abundance table and their classification



https://github.com/hsgweon/pipits

NextITS https://next-its.github.io

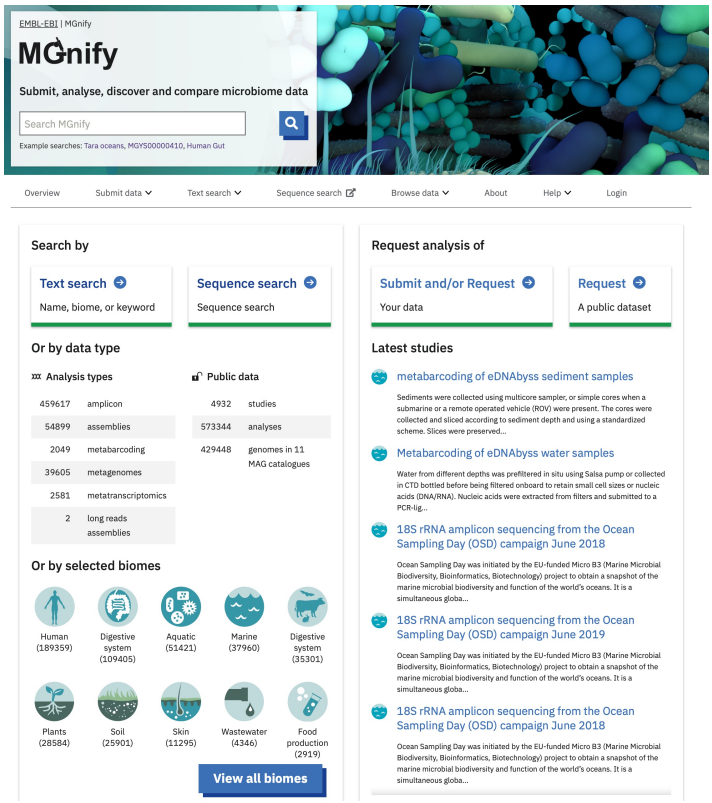
NextITS is an automated pipeline for metabarcoding fungi and other eukaryotes with full-length ITS sequenced with PacBio.
(clustering to OTU)



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Mikryukov V., Anslan S., Tedersoo L. NextITS: a pipeline for metabarcoding fungi and other eukaryotes with full-length ITS sequenced with PacBio. https://github.com/vmikk/NextITS

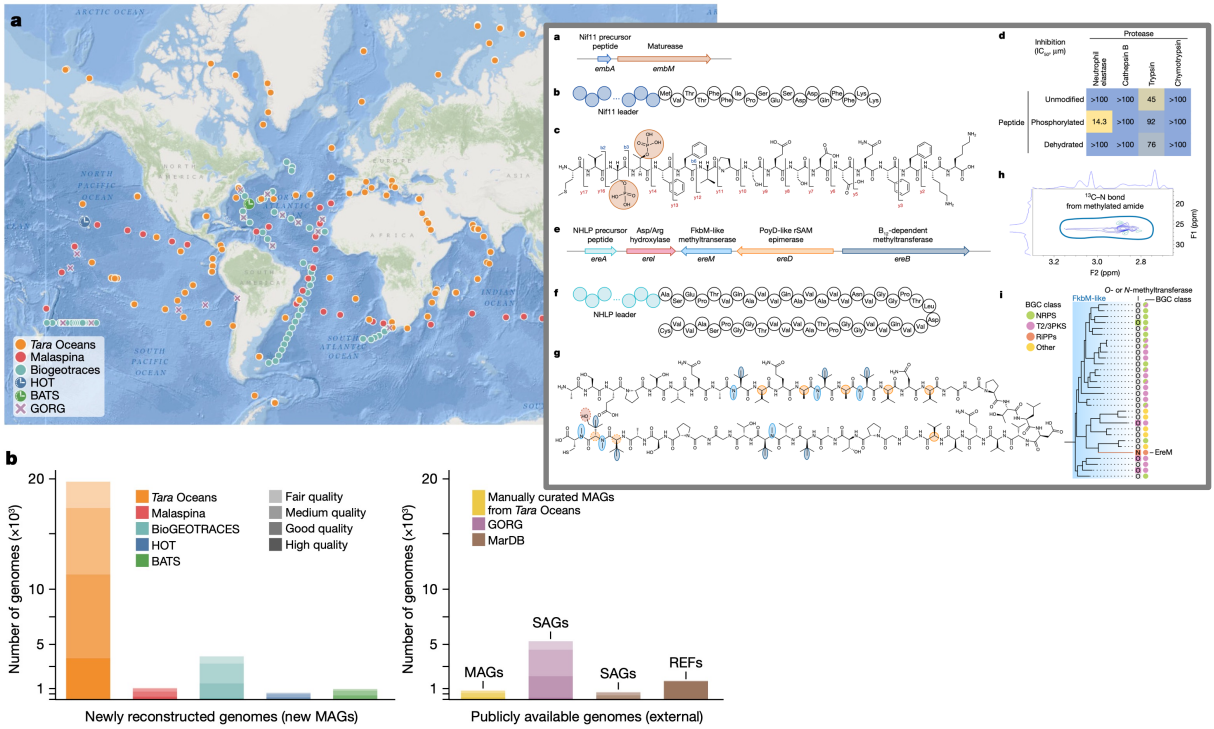
MGnify the database available from EBI



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https://www.ebi.ac.uk/metagenomics/

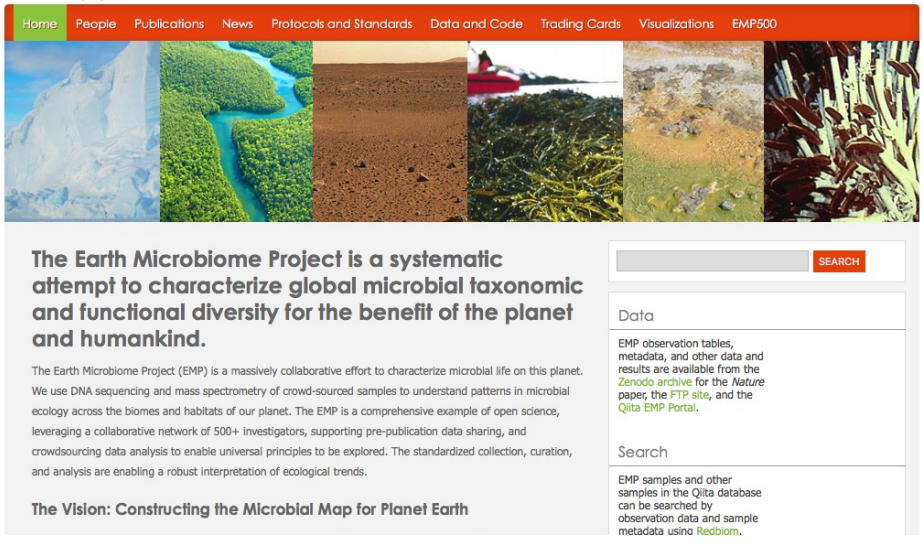
Ocean metagenome & pathways https://microbiomics.io/ocean/



Paoli, L., Ruscheweyh, HJ., Forneris, C.C. et al. Biosynthetic potential of the global ocean microbiome. Nature 607, 111–118 (2022). https://doi.org/10.1038/s41586-022-04862-3

Earth Microbiome Project

The Earth Microbiome Project (EMP) is a systematic attempt to characterize global microbial taxonomic and functional diversity for the benefit of the planet and humankind. Most of the data generated to this point are from 16S rRNA amplicon sequencing, but the project also includes data from 18S and ITS amplicon sequencing, metagenomics, and metabolomics. https://www.earthmicrobiome.org



Summary

Method	Targeted	WGS
Short reads	**Currently standard** Many biaises	**Currently standard** Several pipelines to choose from
Long reads (hifi)	Becomes competitive Many advantages Added throughput by Kinnex	Still expensive But much easier to obtain MAGs
Hybrid (short+long)	Not used in practice	Rarely combined

Summary

Importance of metagenomes for understanding our environment

Two main methods exist: Shotgun vs Targeted
Both methods extract different things from the data
e.g., Taxonomy assignment or functional assignment

Two sequencing techniques compete: short reads vs long reads

Many initiatives and databases gathering data

More details in the coming days !

Thank you for your attention. Questions?

